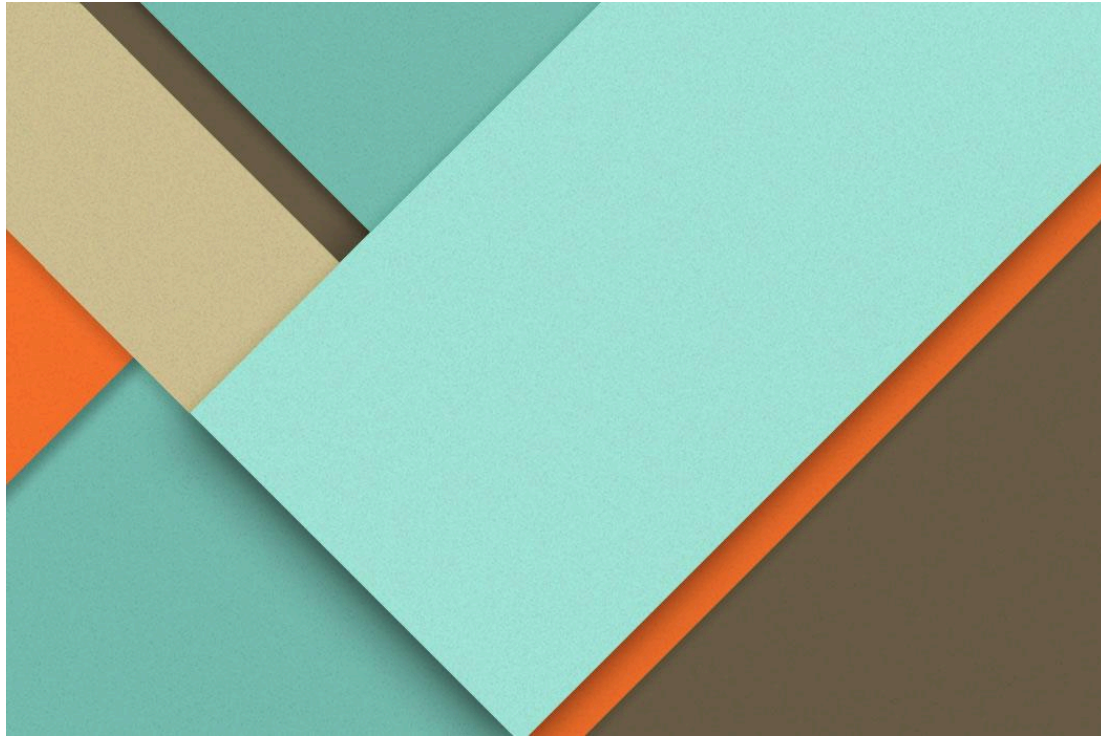




Adventutor

09.14.2025

Group 1 - MAWD 102

Lance Alexis D. Bathan

Alexi Troy C. Indicio

Zandro C. Legaspi

Cye Jay Z. Adecer

James Patrick N. Frias

Khaizzer F. Dalumpines

Yusri C. Ilagan

David Rafael P. Virtusio

Wayne Scott L. Bathan

Issue

1. The issue is that dyslexic learners struggle with reading and processing written information due to difficulties in phonological processing and decoding. This makes learning slower and more challenging, which is the problem our proposal aims to solve. Dyslexia affects an estimated **5–15% of the global population** regardless of language, culture, or country. This makes it a **global learning disability** issue.
2. For people who are poor, dyslexia can lead to amplified negative academic, economic, and emotional outcomes due to fewer resources, less access to specialized support, and greater difficulty in finding skilled employment. Poverty-related stress can worsen existing emotional difficulties like anxiety and depression, and the combination of dyslexia and financial hardship can create a cycle of low educational attainment and limited opportunities.

Why Poverty Worsens These Effects:

Limited Access to Support:

Families with limited financial resources often lack access to specialized dyslexia interventions and educational resources needed to address reading difficulties effectively.

Increased Stressors:

Poverty adds additional layers of stress, which can impair emotional regulation and negatively impact a child's ability to focus on learning.

Lack of Recognition:

In low-resource environments, the specific nature of dyslexia may not be well-understood or recognized by educators, leading to the "incurable" label and further negative impacts.

How Does Adventutor Benefit the Lower Class:

Better access to education:

Students with dyslexia who can't afford to enroll can use Adventutor's hardware as an alternative, since these are easy to print cards that may contain reading practices and challenges that can help them as an alternative for school. Of course our website will still prioritise the student's school learning that is why this is only an alternative.

Improving reading and writing skills:

Adventutor caters to any person, regardless of your status. We strive to help everyone who struggles with dyslexia and by making it accessible, our users can foster their love for learning and improve their reading, writing, creativity, and many more skills.

Increase independence and self-confidence:

By providing alternative ways to access and process information, these tools foster greater independence and reduce reliance on others for reading, writing, and other tasks. Successfully completing tasks, whether academic or professional, with the help of assistive technology can significantly improve a person's self-esteem and motivation.

Effects

Reading starts with spoken language. In early childhood, speaking starts with making simple sounds. As you learn more sounds, you also learn how to use sounds to form words, phrases and sentences. Learning to read involves connecting sounds to different written symbols (letters).

This is where dyslexia enters the picture. It interferes with how your brain uses spoken language to "decode" writing. Your brain has trouble processing what you read, especially breaking words into sounds or relating letters to sounds when reading.

Dyslexia Facts

1. It is estimated that **1 out of 10 people** have dyslexia.
2. The world has roughly 7.8 billion people living it in, where **780 million** of them are dyslexic.
3. 1 out of every 5 students, representing **15% to 20%** of the total, have a learning disability centered on language, with dyslexia being the most common disability.
4. Between **70% to 80%** of kids attending schools with a large minority population have poor reading abilities.
5. 89% of students with an individualized education have problems reading, with **85%** of them having dyslexia.

Source: <https://pubmed.ncbi.nlm.nih.gov/25444408/>

Statistical data / Prevalence

Category	Dyslexia Statistic
Dyslexia Prevalence	~10-20% of people
Percentage of Filipino People With Dyslexia	~5-10%
Most Common Learning Disability	Dyslexia
Men vs. Women Prevalence	~60%, ~40%
Prevalence of Dyslexia Among Grade 1 Students in the Philippines	~13.82%

Percentage Of NASA's Workforce With Dyslexia

~50% (Anecdotal)

Statistical Sources

Dyslexia Prevalence

Source: <https://pmc.ncbi.nlm.nih.gov/articles/PMC8183124/>

Percentage of Filipino People with Dyslexia

Source: <https://philippinedyslexiafoundation.org/index.php/about-dyslexia/>

Most Common Learning Disability

Source: <https://www.discoveryaba.com/statistics/learning-disabilities-prevalence>

Men vs. Women Prevalence

Source: <https://link.springer.com/article/10.1007/s11881-998-0003-8>

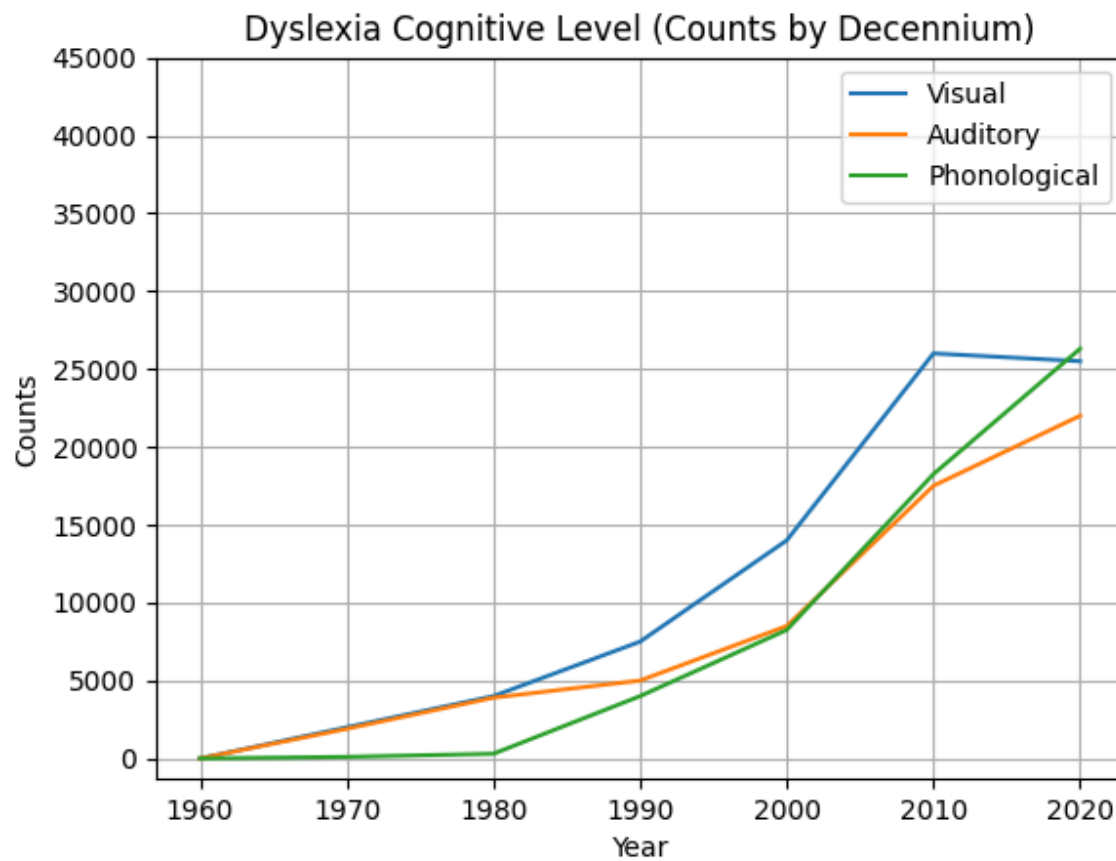
Prevalence of Dyslexia Among Grade 1 Students in the Philippines

Source: <https://www.herdin.ph/index.php?cid=54214&view=research>

Percentage of NASA's Workforce with Dyslexia

Source: <https://solve.mit.edu/solutions/6604>

Visual dyslexia cases maintained the highest absolute counts



Visual:

Year	1960	1970	1980	1990	2000	2010	2020
Visual Count	0	2000	4000	7500	14000	26000	25500

Auditory:

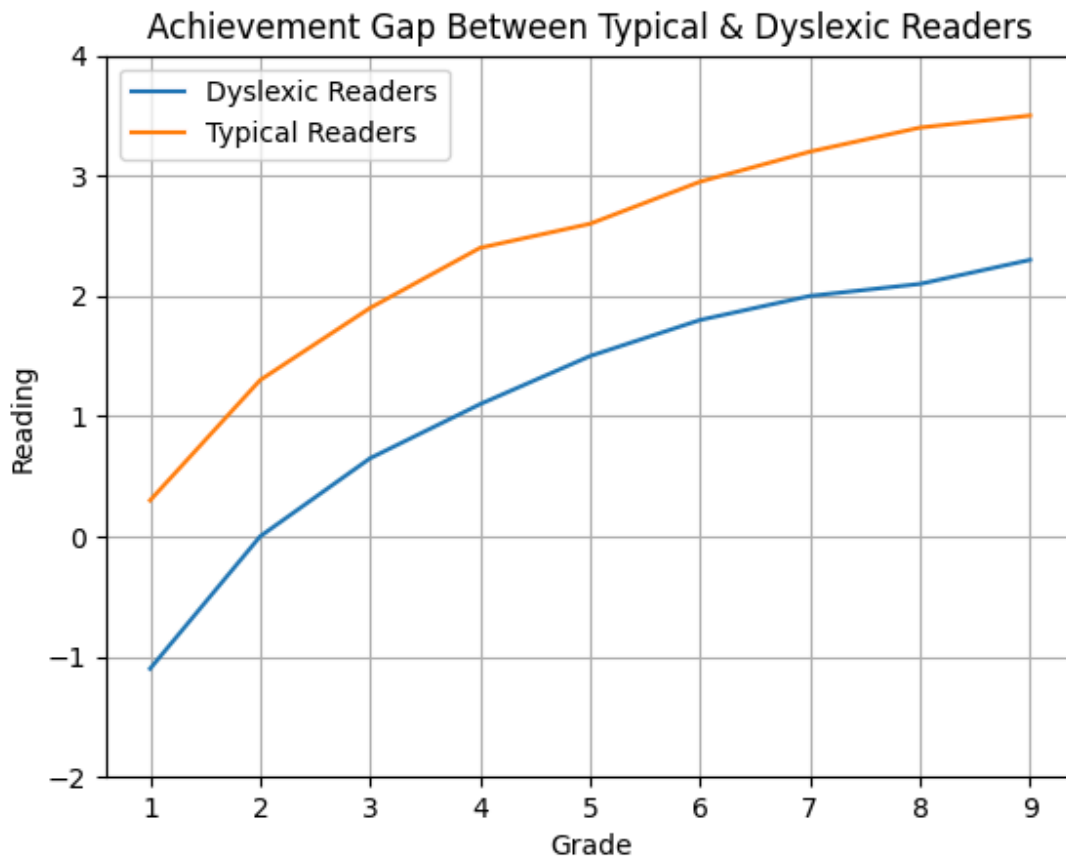
Year	1960	1970	1980	1990	2000	2010	2020
Auditory Count	0	1900	3900	5000	8500	17500	22000

Phonological:

Year	1960	1970	1980	1990	2000	2010	2020
Phonological Count	0	100	300	4000	8250	18250	26300

Source: <https://www.mdpi.com/2076-3425/12/10/1323>

While both groups show improvement across grade levels, typical readers consistently outperform dyslexic readers, and the gap in reading ability widens with grade progression



Dyslexics:

Grade	1.0	2.0	3.0	4.0	5.0	6.0	7.0	8.0	9.0
Reading	-1.1	0.0	0.65	1.1	1.5	1.8	2.0	2.1	2.3

Typicals:

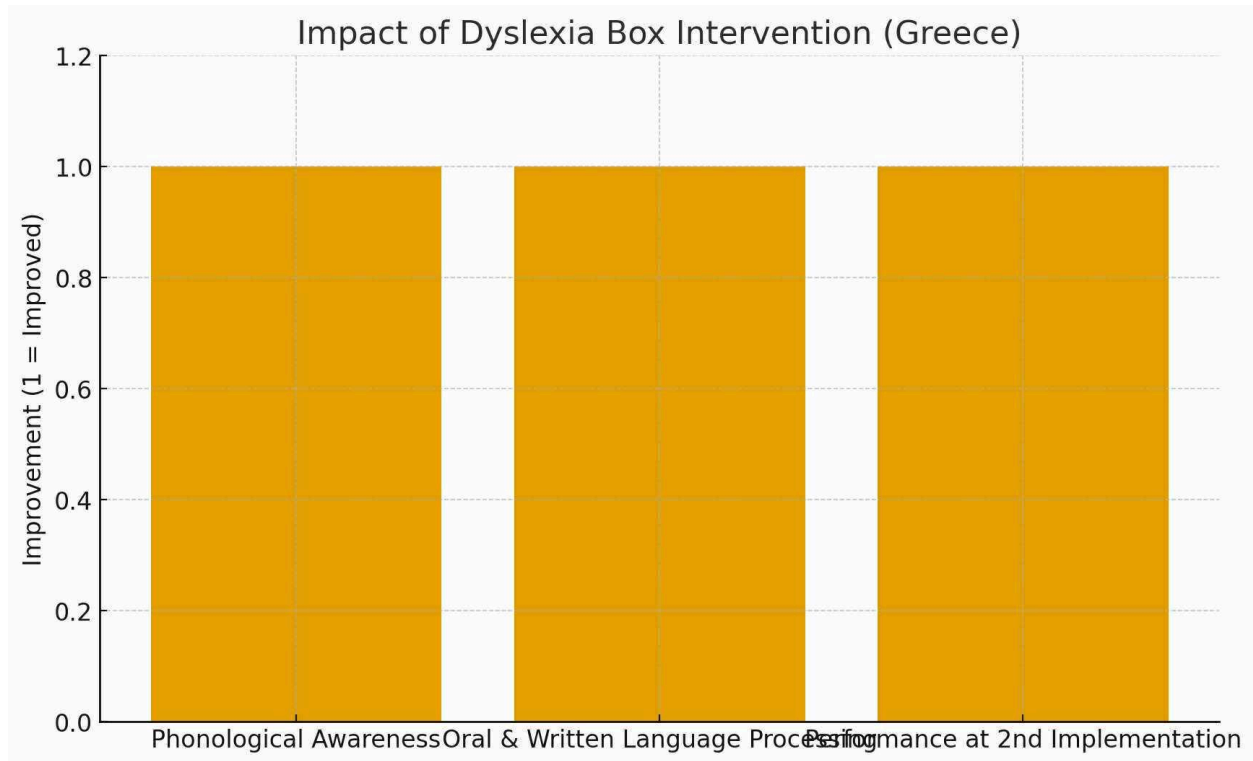
Grade	1.0	2.0	3.0	4.0	5.0	6.0	7.0	8.0	9.0
Reading	0.3	1.3	1.9	2.4	2.6	2.95	3.2	3.4	3.5

Source: [https://www.jpeds.com/article/S0022-3476\(15\)00823-9/pdf](https://www.jpeds.com/article/S0022-3476(15)00823-9/pdf)

Case Studies Sources

The Impact of Dyslexia Box (Greece)

Source: <https://www.mdpi.com/2227-7102/13/5/486>



Barton Intervention Program (Iran)

Source:

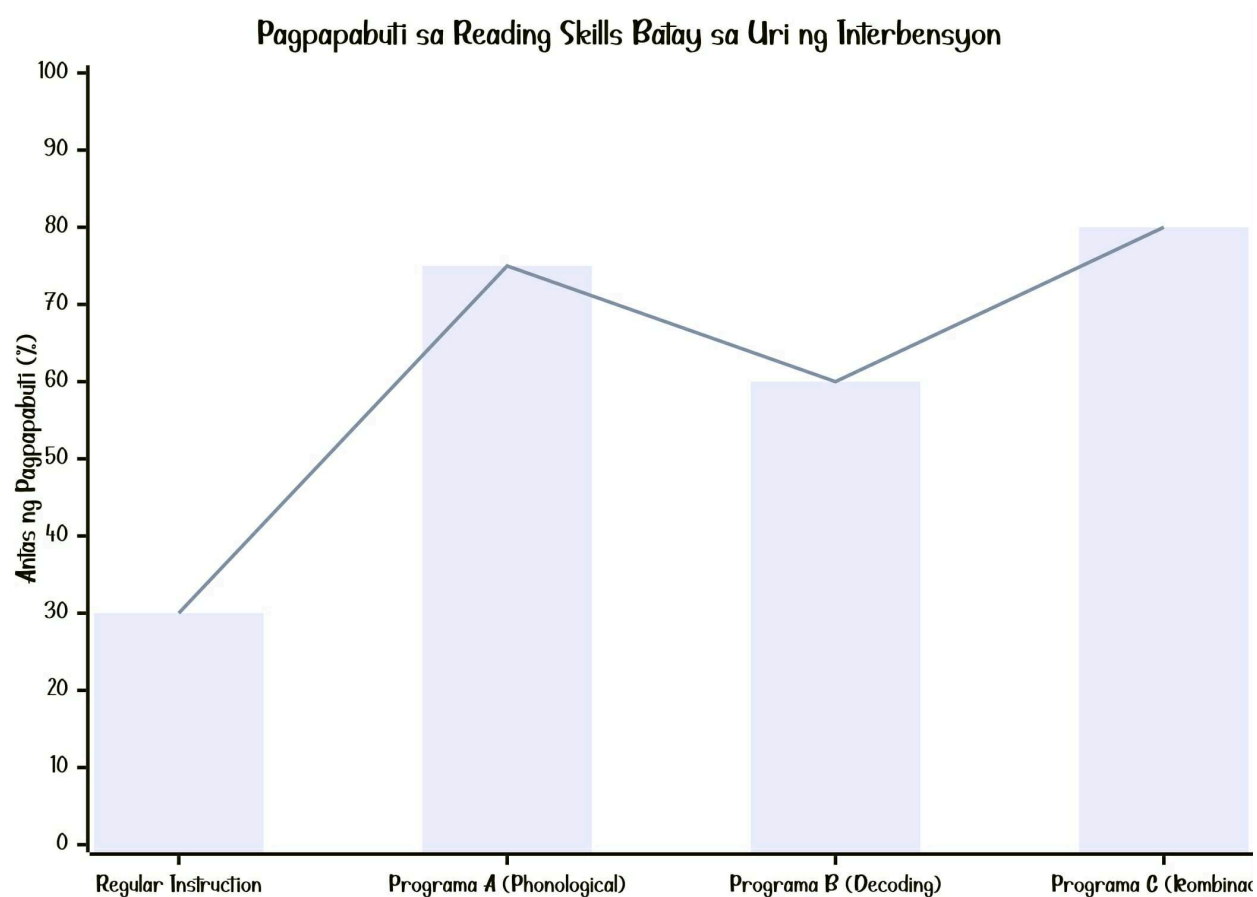
https://journals.lww.com/jehp/fulltext/2019/08000/the_effectiveness_of_an_intervention_program.167.aspx

Sustained Reading & Spelling Intervention (Netherlands / Belgium region)

Source: <https://onlinelibrary.wiley.com/doi/full/10.1002/dys.1614>

Computer-assisted Interventions

Source: <https://pubmed.ncbi.nlm.nih.gov/23338977/>



Case Studies

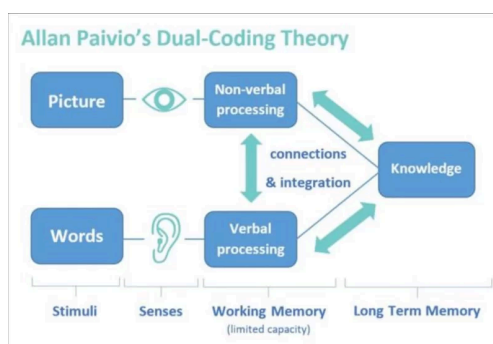
Study	What was done / sample	Findings
The Impact of Dyslexia Box (Greece)	13 students with dyslexia and other language difficulties; intervention over 10 weeks, 32 activities targeting phonological awareness, reading, writing.	Students improved in phonological awareness and in processing of oral and written language; better performance at second implementation. Shows that a structured intervention tool even in integration (inclusive) classes can help.

Barton Intervention Program (Iran)	68 dyslexic students, grades 2-3; experimental vs control; 10-week Barton programme; reading fluency tested pre- and post-intervention.	Statistically significant improvements in reading fluency for the experimental group vs control. Suggests that explicit, structured reading interventions help.
Sustained Reading & Spelling Intervention (Netherlands / Belgium region)	Children with dyslexia are given a two-phase intervention — first focusing on accuracy (declarative), then fluency (procedural + declarative) over multiple sessions.	Children with dyslexia improved in word and pseudoword decoding, spelling, etc., more so after both phases. However even after intervention, typical readers still outperformed them; shows that early, sustained, multi-component work helps but does not necessarily “normalize” entirely.
Computer-assisted interventions	Grade-2 children with reading difficulties; used various computerized training programs over a year; comparison with regular instruction.	All groups improved reading skills; some programs were more effective for certain skills (e.g. phonological/word decoding) than others. Suggested technology can assist, but it's not a silver bullet.

Overview

This is a learning platform dedicated for individuals with dyslexia, we plan to focus on **image-based learning**, intuitive visualizations, and engaging activities. It features a system where users earn coins for completing tasks, making learning interactive and motivating.

Image-based learning



According to the theory above

It says here that the brain processes verbal and visual information separately, meaning that listening to information is not the same as reading or seeing it.

Source:

[https://www.researchgate.net/publication/225249172 Dual Coding Theory and Education](https://www.researchgate.net/publication/225249172_Dual_Coding_Theory_and_Education)

Why is this important?

What's the main problem for people who suffer from dyslexia? It is the difficulty in reading, because they see letters and characters differently compared to people without dyslexia, which makes reading challenging for them.

This can be addressed or improved by using an image-based learning system. When information is presented both visually and verbally, it creates two mental representations, making it easier to understand and remember.

Experiments show that people are more likely to remember images than words. Individuals with dyslexia tend to recall visual information better than text because visuals reduce cognitive load and engage pattern recognition, making concepts easier to understand, retain, and recall.

Source: <https://www.dyslexicadvantage.org/visual-learning-premium/>

This article says that a survey with 12,291 members was held and the results show that 70% of dyslexic people strongly agreed that they prefer diagrams and pictures.

Adventutor Web**1. Home Screen****Tag lines:**

"Where learning becomes a journey of wonder"

"A conquest for knowledge"

"Turn study time into Adventure time"

Shows today's quests: (Example)

✓ Play 1 reading mini-game

✓ Focus for 5 minutes

✓ 5 minute writing practice

- Once they complete tasks, a progress bar fills up and grows their virtual pet/character.
- If the user stays inactive for 3 days, the pet's health deteriorates. This feature promotes consistency of use for the user. (like Po)



2. Learning Zone

- ! Mini-games for study (drag-and-drop, puzzles, flashcards)
- ! Adaptive reading tools
- ! Dyslexia-friendly font
- ! Read-aloud text (with highlighting)
- ! Tap to hear words

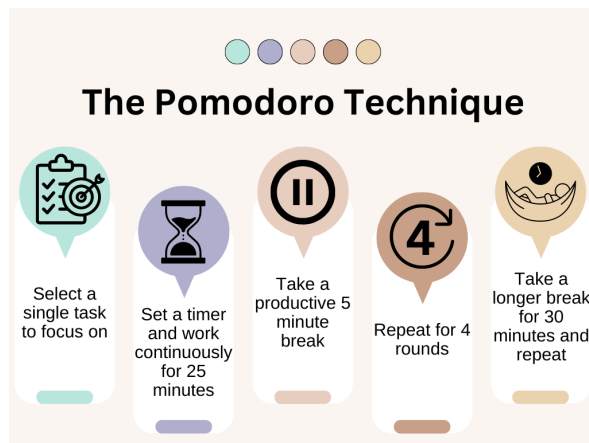


3. Focus Zone

- ! "Pomodoro for kids"
- ! 5–10 min timer with animations.
- ! Rewards when they finish (pet gets food, coins, etc.).

What is Pomodoro?

Pomodoro technique is a time management method that breaks down work into 25 minute intervals.



4. Calm Zone / Break Zone

- Simple breathing exercise mini-games.
- Emotion check-in ("How do you feel? 😊 😐 😞").

5. Pet / Character Zone

- A pet/robot/plant that evolves when kids complete daily tasks.
- Different skins/looks unlocked by streaks (who doesn't love uniqueness).
- Promotes consistency, not just correctness.

6. Parent / Teacher Dashboard

- See progress (time focused, reading levels, mood check ups).

! Get suggestions (“Child prefers audio prompts – try more listen-and-repeat games”).

! Mag assign ng custom “quests” (like homework tasks).



Statement of the Problem

- **How can the platform adapt to learning speeds and needs?**

Learners using Adventutor with adaptive learning features will show higher reading speed, improved spelling accuracy, and increased engagement compared to learners using a non-adaptive platform.

- **What features make learning more intuitive and less frustrating for people with literacy difficulties?**

The inclusion of features such as multisensory cues, dyslexia-friendly fonts, and adaptive exercises increases learners' perceived intuitiveness and reduces frustration compared to a standard learning interface.

- **How can the platform serve users who are not formally diagnosed but still struggle with literacy?**

Learners using an adaptive learning system that adjusts to their prior knowledge and performance will demonstrate more consistent improvements in reading accuracy, spelling, and engagement compared to learners using a non-adaptive system.

Goals and Objectives

- **Enhance Focus and Time Management**

Integrate Pomodoro timers and focus-based na mini-games to help learners of any age manage study time effectively.

- **Gamify Learning for Motivation**

Use quests, streaks, coins, and a virtual pet system to encourage consistent study habits rather than focusing only on grades.

Proposed Solution

To address the challenges of illiteracy, our project proposes a learning platform that emphasizes image-based instruction, visual imagery, and engaging activities. By transforming traditional text-heavy lessons into interactive visual content, the platform aims to make learning easier to understand, easier to retain, increase enjoyability, and adaptable to all users.

Target Users of Beneficiaries

Adventutor is a website focused on helping individuals who suffer from dyslexia and other reading-related challenges. Our target users include people of all ages who were born as slow readers, poor spellers, or struggling writers, many of whom may have never received the proper support in traditional educational settings.

Planned Technology or Tools

Adventutor will use AR (Augmented Reality) Flash Cards that will feature QR codes that our user can scan. There will be different types of cards:

1. Interactive Cards

- These cards feature 3D versions of the images in their card kung saan pwede na makipag interact and user. These cards are more focused on expanding understanding among our users about certain topics only limited to photos. After scanning, they can move around the model to explore.

2. Puzzle Cards

- These cards are minigames and puzzles that our user can solve or play to receive various rewards like pets, and more customisation.

3. Mythical Cards

- These are rare cards that may contain challenges that the users can complete that will give them rare pets. These cards are only one scan per card.



Comparison Tools

1. Rewordify

Pros:

Simplifies complex sentences into easier-to-read language.

Highlights key vocabulary to aid comprehension.

Free to use, so accessible to anyone.

Cons:

Cannot read text aloud; auditory support absent.

Over-simplifies content, losing nuanced meaning.

Works only with text; PDFs/images not fully supported.

Affordability: Free.

2. Scrible

Pros:

Allows highlighting and annotation to organize thoughts visually.

Helps users keep track of key information in long texts.

Can collaborate with others, helpful for educational settings.

Cons:

No built-in text-to-speech for reading aloud.

Paid version needed for full features.

Can be complex to learn for students with reading difficulties.

Affordability: Free basic plan; premium subscription for full features.

3. Google Voice Typing (in Google Docs)

Pros:

Converts speech to text, helping those who struggle with writing.

Integrated with Google Docs, so no extra apps needed.

Free for all Google account users.

Cons:

Does not help with reading text; only writing assistance.

Requires accurate pronunciation; may misinterpret speech.

Formatting and punctuation must be corrected manually.

Affordability: Free.

4. Microsoft Immersive Reader

Pros:

Reads text aloud, highlighting words as it goes.

Offers line focus, syllable splitting, and font adjustments for easier reading.

Free to use on supported platforms.

Cons:

Limited platform compatibility; not all websites/apps supported.

Voices cannot always be customized.

Does not provide advanced comprehension aids beyond reading support.

Affordability: Free.

5. Grammarly

Pros:

Helps identify spelling/grammar errors for struggling writers.

Word suggestions can improve clarity for people with reading challenges.

Browser extension available for real-time assistance.

Cons:

No text-to-speech; does not help with reading difficulties.

Advanced grammar suggestions require paid plan.

May be overwhelming for users with severe dyslexia.

Affordability: Free basic version; premium subscription for advanced features.

6. Quizlet / Cram

Pros:

Flashcards support memorization of vocabulary/terms.

Some flashcards include audio, aiding auditory learners.

Simple, intuitive interface for struggling readers.

Cons:

Audio not available for all cards.

Over-simplifies content; may not support comprehension of complex texts.

Premium needed for full features like offline access.

Affordability: Free basic version; optional premium subscription.

7. BeeLine Reader

Pros:

Uses gradient text colors to guide eyes, helping focus and reduce reading fatigue.

Helpful for ADHD or dyslexic readers.

Browser extension is easy to activate on web pages.

Cons:

Full features require subscription.

Only works with digital text, not printed materials.

Some websites may not be compatible.

Affordability: Free trial; subscription required for unlimited use.

8. Helperbird

Pros:

Provides dyslexia-friendly fonts, text-to-speech, and background color options.

Highly customizable to individual reading needs.

Works directly in the browser for web accessibility.

Cons:

Some advanced features require a paid plan.

Browser-only; cannot help with offline materials.

Interface may take time to set up for beginners

Affordability: Free basic plan; Subscription for full suite

What they lack, and the opportunities:

-Many existing tools for dyslexic students or individuals who struggle with reading provide valuable support, but they each tend to focus on only one or two aspects of the learning process. Tools like Rewordify simplify text to make reading easier, Microsoft Immersive Reader and Voice Dream Reader offer text-to-speech and visual aids, and apps like Grammarly or Quizlet improve writing and memorization skills. Browser extensions like BeeLine Reader and Helperbird add dyslexia-friendly fonts, line guides, or color overlays to make text more accessible. While these features are all helpful individually, none of these tools combine all the necessary elements—text simplification, auditory support, visual customization, interactive engagement, and motivation—into a single, cohesive platform.

Many require users to switch between multiple apps, learn complex interfaces, or pay for premium features to access the full benefits. Additionally, most lack gamification or fun elements, which can be critical for keeping students engaged and motivated over time. Offline access, multi-format support, and progress tracking are often limited or absent, leaving gaps in usability for real-world learning situations. Essentially, while these tools each address a specific need, there is a clear opportunity for a solution that integrates all essential reading supports into one accessible, engaging, and motivating platform, which is exactly what our app aims to provide.

Team Roles

Project Leader:

Lance Alexis D. Bathan

Hardware Builder:

Lance Alexis D. Bathan

James Patrick N. Frias

David Rafael P. Virtusio

Programmer:

Lance Alexis D. Bathan

James Patrick N. Frias

David Rafael P. Virtusio

Alexi Troy C. Indicio

UI Designer:

Lance Alexis D. Bathan

James Patrick N. Frias

David Rafael P. Virtusio

Khaizzer F. Dalumpines

Zandro C. Legaspi

Wayne Scott L. Bathan

Yusri C. Ilagan

Cyejay C. Adecera

Documenter:

Lance Alexis D. Bathan

David Rafael P. Virtusio

Khaizzer F. Dalumpines